

UV Catalysis, Cyanotype Photography, and Sunscreens W

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Ultraviolet radiation can be an effective catalyst for a wide range of chemical reactions. It is often discussed in the context of being damaging to living organisms because it can initiate mutations in DNA that may result in skin cancer and other health problems. Most of the sun's UV rays are filtered by the gases in our atmosphere: the very-high-energy UV light is absorbed by oxygen and the intermediate-energy UV light is absorbed by ozone in the upper atmosphere. There is much concern that chlorofluorocarbons (CFCs) and other anthropogenic pollutants released into the atmosphere during the past half century may be destroying the protective ozone layer, because UV activation of the CFCs initiates free radical reactions that destroy the ozone. Sunscreens are sold to protect us from the damaging effects of the UV radiation that manages to penetrate the atmosphere and strike our exposed bodies. A spectrophotometric approach to UV absorption by sunscreens and sunglasses was recently discussed in this *Journal* (1). Courses in chemistry for non-science majors often discuss these reactions (2), although few laboratory experiments are used in such courses to illustrate the catalytic effect of light, especially UV light, on chemical reactions.

The chemical processes for UV-activated photographic image processing have been known since the mid-19th century, and various modifications in recent years have improved methods for photographers. The cyanotype process, invented by Sir John Herschel in the early 1840s (3), was one of the more significant discoveries in photography because it resulted in a chemically stable, permanent print that could be used to make archival prints, was relatively nontoxic, and led to several other photographic processes. Commercial paper for the cyanotype process was available in the 1870s for marketing to engineers, architects, and draftsmen for copying drawings (blueprints). This technique is still in use because it is less expensive than newer technologies. Cyanotype printing became popular among amateur photographers toward the end of the 19th century because of its simplicity and low cost, although the bright blue color prevented its adoption by the "serious" photographer (4, 5).

In the cyanotype process the paper is sensitized with an aqueous mixture of ferric ammonium citrate and potassium ferricyanide. After drying, exposure to ultraviolet light causes some reduction of the Fe(III) (ferric) salts to Fe(II) (ferrous) with citrate as the electron donor. The Fe(II) ion complexes with ferricyanide ion, with subsequent electron transfer, to form insoluble ferric ferrocyanide—iron(III) hexacyanoferrate(II), or Prussian blue (6). Because the image appears as the paper is exposed to UV light (without having to develop it), this is known as a printing-out process. After sufficient exposure, the paper is washed in water to remove the soluble unexposed salts. Upon drying, the final image darkens as a result of either slow oxidation in air or some changes in iron coordination with loss of water. This blue

pigment is practically insoluble in water and has been used for printing ink, paint pigment, typewriter ribbon, and carbon paper (7). Treatment with oxidants such as hydrogen peroxide or potassium dichromate produces a darker blue (almost black) image. Washing in aqueous ammonia solution results in some loss of the color (the pigment dissolves in the solution and so washes out of the paper permanently). For a complete discussion of the controversy regarding structure of the blue cyano complexes of iron and their alkaline stability see Holtzman (8).

The cyanotype process is convenient to use in the chemistry laboratory or for a demonstration because it is possible to work with this photographic material in visible light in a room with windows or a room lit with incandescent light, although direct sunlight and standard fluorescent lights have sufficient ultraviolet radiation to catalyze the chemical changes that take place in formation of the image on paper. This often prompts students to ask why this paper can be handled in visible light, whereas anyone who has taken a beginning course in photography knows that this light will destroy commercial (silver gelatin) photographic paper. The photodynamic range of photographic papers, as well as other devices such as light meters in cameras and semiconductors, can be discussed in this context.

The chemical reactions taking place in the cyanotype process are not well understood, beginning with the uncertain structure for ammonium ferric citrate. The structure of insoluble Prussian blue is not known in detail, but is most likely $\text{Fe}^{\text{III}}_4[\text{Fe}^{\text{II}}(\text{CN})_6]_3 \cdot n\text{H}_2\text{O}$ (9). The so-called soluble Prussian blue, which is not water soluble but is more finely dispersed in water, has the formula $\text{KFe}^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot n\text{H}_2\text{O}$ and is also likely to be one of the products of this reaction (10). Ammonium ferric citrate is the photosensitive component in the sensitizer. Photoactivation results in reduction of Fe(III) to Fe(II), with oxidation of citrate to liberate CO_2 , although the oxidation products have not been identified. A subsequent electron transfer from the Fe(II) ion to hexacyanoferrate(III) results in the more stable iron(III) hexacyanoferrate(II) product with variable amounts of hydration. Because of the variability in chemical structure of Prussian blue, the chemical reaction taking place upon treatment with hydrogen peroxide or other oxidants (e.g., ammonium dichromate) to darken the color is not clear.

It is most convenient to make contact prints, so the print will be the same size as the negative used to make the photographic image. Large format (5 × 7 in. or 13 × 18 cm) black and white negatives can be obtained from professional photographers or in a college photography department. It is possible to use a wide variety of objects that will produce negative images on the paper (stencil for lettering, keys, sunglasses, jewelry, etc.) or create negatives by drawing with black marker on acetate sheets or cutting out images (silhouettes)

in paper. We have extended the use of this simple photographic technique to test the efficacy of sunscreens with different SPF (sun protection factor) ratings.

Materials Needed

Artist's watercolor or bristol paper, ammonium ferric citrate (available in green or brown forms—the green form works best), potassium ferricyanide, 3% hydrogen peroxide solution (available from a pharmacy), 0.2 M aqueous ammonia solution (or dilute clear household ammonia), trays or plastic wash basins for hydrogen peroxide and aqueous ammonia solutions, a beaker to mix the sensitizer solution, paint brush (bristle or inexpensive sponge type, must fit inside beaker), blow dryer, negatives or objects to produce an image on the photographic paper, clear acetate sheets, samples of skin lotion with and without sunscreen, and direct sunlight or ultraviolet light source (detailed instructions for building an inexpensive light box are included with material online).^W

Procedure

It is possible to work under incandescent light or daylight coming through a window, but fluorescent lights should be turned off and direct sunlight should be avoided. Mix equal amounts of ammonium ferric citrate solution and potassium ferricyanide solution in a beaker or other wide-mouth container to make the sensitizer solution (5–10 mL of sensitizer solution is sufficient for several sheets of paper). Paint or spread this solution as evenly as possible on the paper (preferably good-quality artist's paper). Dry the paper with a hair dryer or in a warm oven. Make sure the treated paper is completely dry before placing the negative on it, or the negative may be permanently damaged by the sensitizer chemicals. If using a photographic negative, place the negative with glossy side up on top of one piece of the completely dry treated paper and place it under glass in the light box for UV light exposure (6–8 minutes in the UV light box described) or in direct sunlight. The time needed for exposure to sunlight will depend on latitude, season, time of day, and weather conditions, but the image appears as the paper is exposed, so it is easy to judge the exposure time. Objects such as leaves or other botanical materials can be used in lieu of a large negative. It is advisable to place a sheet of glass over the negative and sensitized paper to keep them flat. It is possible to see the image appear on the paper as it is exposed to UV light. When the image looks medium to dark blue, it has probably been exposed long enough.

To test sunscreens for their ability to block UV rays, apply a thin film of lotion to a sheet of glass or clear acetate using a glass microscope slide or flat piece of plastic to spread the lotion and get a uniform thickness of the film (a complete description of this procedure is given online).^W Place the glass or acetate sheet with lotion samples (lotion side up) on a piece of completely dry sensitizer-treated paper. Allow 6 to 8 minutes light exposure in the box or expose to sunlight until the unprotected areas have turned medium to dark blue.

After sufficient exposure, remove the negative or objects from the paper and wash the paper in a tray with running tap water for about 5 minutes. This should give a good permanent image (Prussian blue). For a darker, almost black image,

rinse the photo briefly (10 seconds) in a dilute hydrogen peroxide solution (3% hydrogen peroxide from a pharmacy is suitable), then rinse for several minutes in running tap water.

If the paper was exposed too long to the UV light, giving too much pigment density, it is possible to produce a lighter image (much of the color will be washed out). Place the photo in dilute aqueous ammonia solution for a few seconds, then rinse well with running tap water. The paper can be treated with hydrogen peroxide after rinsing with aqueous ammonia to get less pigment density and a darker tone. The paper should be rinsed briefly with tap water after each treatment to avoid contaminating the trays of chemicals. Wash the paper thoroughly and dry when finished.

Discussion

This experiment illustrates the catalytic effect of ultraviolet radiation on some chemical reactions, in contrast to visible light, which is less effective because of its lower energy. It is inexpensive to perform if sunlight is used as the source of UV light. Students enjoy the experiment because they can be creative in designing an image for the photograph, and they are often fascinated by the fact that it is possible to work in visible light and they can make their own sensitized paper. The chemicals are not dangerous to use, although potassium ferricyanide and aqueous ammonia should be used with care. The experiment is useful in introducing not only the properties of UV light, but also the chemistry of photography. Students and instructors may want to look into other photographic processes.

This experiment is used in a chemistry course for nonscience majors, for which the text *Chemistry in Context* (2) is used.

Note

^WSupplementary materials for this article are available on JCE Online at <http://jchemed.chem.wisc.edu/Journal/issues/1999/Sep/abs1199.html>. A classroom activity based on this article appears in this issue on pages 1216A–B.

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